Coding Problems in Java - Set 1

1. Print Numbers from 1 to N

Write a program that takes an integer N as input and prints numbers from 1 to N using a loop.

Example:

```
Input: 5
Output: 1 2 3 4 5
```

2. Print Even Numbers up to N

Write a program to print all even numbers from 1 to N.

Example:

```
Input: 10
Output: 2 4 6 8 10
```

3. Calculate the Sum of First N Natural Numbers

Write a program to calculate the sum of numbers from 1 to N.

Example:

```
Input: 10

Output: 55
```

4. Reverse a Number

Write a program to reverse the digits of an integer.

Example:

```
Input: 1234
Output: 4321
```

5. Find Factorial of a Number

Write a program to compute the factorial of a given number N.

Example:

```
Input: 5
Output: 120  (5! = 5×4×3×2×1)
```

6. Check if a Number is Prime

Write a program to check whether a number N is prime.

Example:

```
Input: 7
Output: Prime Number
```

7. Find the Sum of Digits of a Number

Write a program to compute the sum of the digits of a number.

Example:

```
Input: 123
Output: 6  (1+2+3)
```

8. Fibonacci Series up to N Terms

Write a program to print the Fibonacci series up to N terms.

Example (n = 7):

```
0 1 1 2 3 5 8
```

9. Print a Star Pattern

Write a program to print a right-angled triangle pattern.

Example (n = 4):

```
*
**
***
****
```

10. Find the Greatest Common Divisor (GCD) using the Euclidean Method.

Write a program to find the GCD (Greatest Common Divisor) of two numbers.

Example:

```
Input: 20, 30
Output: 10
```

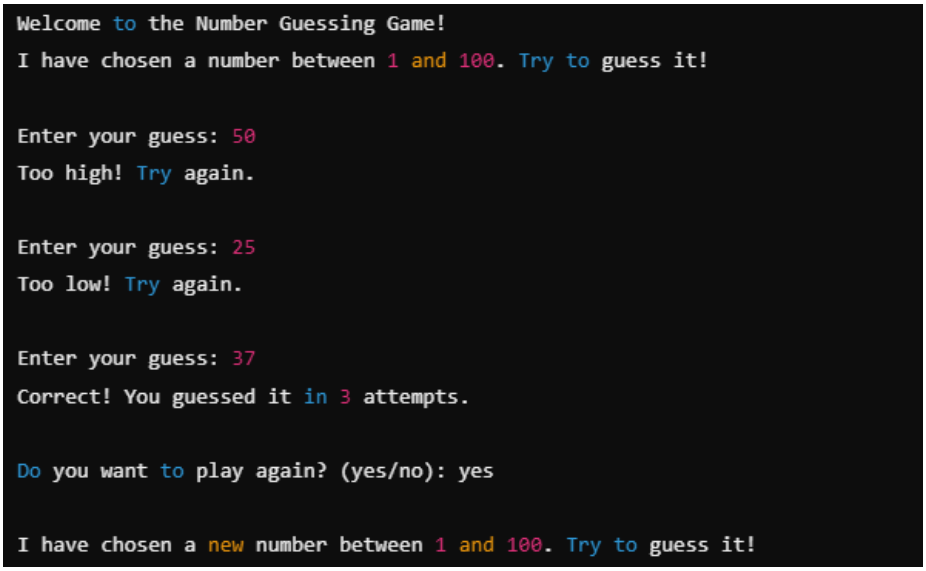
11. Major Task: "Guess the Secret Number"

Problem Statement: Create a Java program where the computer randomly selects a number between 1 and 100. The player has to guess the number. After each guess, the program should give hints:

- **"Too High" if the guess is higher than the secret number.**
- **"Too Low" if the guess is lower than the secret number.**
- **"Correct!" if the guess matches the number.**

Key Concepts Covered:

- **while loops and do-while loops**
- **if-else conditions**
- **Random number generation (Math.random())**
- **User input with Scanner**
- **Basic error handling with try-catch (optional challenge)**



```java
import java.util.Scanner;
import java.util.Random;

public class NumberGuessingGame {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Random rand = new Random();
        String playAgain;

        System.out.println("Welcome to the Number Guessing Game!");

        do {
            int secretNumber = rand.nextInt(100) + 1;  // Random number between 1 and 100
            int guess = 0;
            int attempts = 0;

            System.out.println("I have chosen a number between 1 and 100. Try to guess it!");

            // Loop until the correct guess
            while (guess != secretNumber) {
                System.out.print("Enter your guess: ");

                // Input validation
                if (sc.hasNextInt()) {
                    guess = sc.nextInt();
                    attempts++;

                    if (guess < 1 || guess > 100) {
                        System.out.println("Please enter a number between 1 and 100.");
                        continue;
                    }

                    if (guess < secretNumber) {
                        System.out.println("Too low! Try again.");
                    } else if (guess > secretNumber) {
                        System.out.println("Too high! Try again.");
                    } else {
                        System.out.println("Correct! You guessed it in " + attempts + " atte
                    }
                } else {
                    System.out.println("Invalid input! Please enter an integer.");
                    sc.next();  // Clear the invalid input
                }
            }
```

```
            // Ask if the player wants to play again
            System.out.print("Do you want to play again? (yes/no): ");
            playAgain = sc.next();

        } while (playAgain.equalsIgnoreCase("yes"));

        System.out.println("Thanks for playing!");
        sc.close();
    }
}
```